

Lancaster University College
at Beijing Jiaotong University

# Digital Signal Processing

Zhou Hang

# Introduction



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- Why learn digital signal processing?
- Curriculum System for Electronic Information Majors
- Main Contents of Digital Signal Processing
- Teaching Objectives of Digital Signal Processing
- Teaching Resources of Digital Signal Processing

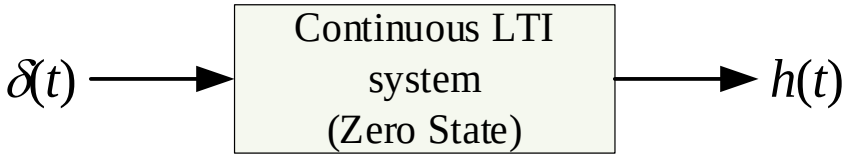
# A Review of the Course "Signals and Systems"

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| Signal Representation | | Mechanism of S&S | | System Description |
|---|---|---|---|---|
| Time Domain | $x(t) \propto \delta(t)$ | $x(t)$ → LTI → | $y(t)= x(t)* h(t)$ | $h(t)$ |
| | $x[k] \propto \delta[k]$ | $x[k]$ → LTI → | $y[k]= x[k]* h[k]$ | $h[k]$ |
| Frequency Domain | $x(t) \propto e^{j\omega t}$ | $X(j\omega)$ → LTI → | $Y(j\omega)= X(j\omega)\, H(j\omega)$ | $H(j\omega)$ |
| | $x[k] \propto e^{j\Omega k}$ | $X(e^{j\Omega})$ → LTI → | $Y(e^{j\Omega})=X(e^{j\Omega})H(e^{j\Omega})$ | $H(e^{j\Omega})$ |
| Complex Frequency Domain | $x(t) \propto e^{st}$ | $X(s)$ → LTI → | $Y(s)= X(s)\, H(s)$ | $H(s)$ |
| | $x[k] \propto z^{k}$ | $X(z)$ → LTI → | $Y(z)=X(z)H(z)$ | $H(z)$ |

# Time-domain Analysis of Signals and Systems





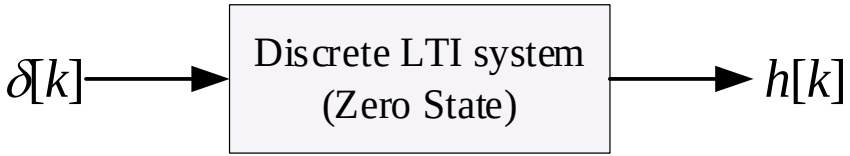

$$x(t) = \lim_{\Delta\tau \to 0} \sum_{k=-\infty}^{\infty} x(k\Delta\tau)\, \delta(t - k\Delta\tau)\, \Delta\tau$$

$$y(t) = \lim_{\Delta\tau \to 0} \sum_{k=-\infty}^{+\infty} x(k\Delta\tau) \cdot\ h(t - k\Delta\tau)\, \Delta\tau$$

$$= \int_{-\infty}^{+\infty} x(\tau) h(t-\tau) \mathrm{d}\tau \ = x(t) * h(t)$$

# Time-domain Analysis of Signals and Systems






$$x[k] = \sum_{n=-\infty}^{\infty} x[n]\,\delta[k-n]$$

$$y[k] = \sum_{n=-\infty}^{\infty} x[n]\,h[k-n] = x[k] * h[k]$$

# Frequency-Domain Analysis of Signals and Systems






$$e^{j\omega t} * h(t) = \int_{-\infty}^{\infty} h(\tau) e^{j\omega(t-\tau)} d\tau = e^{j\omega t} H(j\omega)$$

$$H(j\omega) = \int_{-\infty}^{\infty} h(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} \, d\omega$$

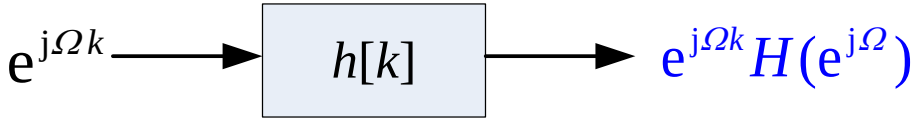
$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \, H(j\omega) e^{j\omega t} d\omega$$

$$Y(j\omega) = H(j\omega) X(j\omega)$$

# Frequency-Domain Analysis of Signals and Systems






$$e^{j\Omega k} * h[k] = \sum_{n=-\infty}^{\infty} h[n]e^{j\Omega(k-n)} = e^{j\Omega k}H(e^{j\Omega})$$

$$H(e^{j\Omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\Omega k}$$

$$x[k] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\Omega})e^{j\Omega k}\,d\Omega$$

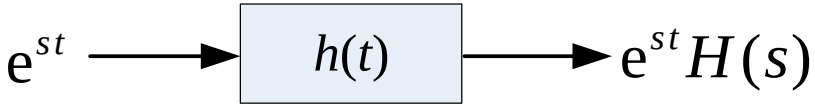
$$X(e^{j\Omega}) = \sum_{k=-\infty}^{\infty} x[k]e^{-j\Omega k}$$

$$y[k] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\Omega})H(e^{j\Omega})e^{j\Omega k}\,d\Omega$$

$$Y(e^{j\Omega}) = H(e^{j\Omega})X(e^{j\Omega})$$

# Complex frequency domain analysis of signals and systems






$$e^{st} * h(t) = \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau = e^{st} H(s)$$

$$H(s) = \int_{-\infty}^{\infty} h(t) e^{-st} dt$$

$$x(t) = \frac{1}{2\pi j} \int_{\sigma-\infty}^{\sigma+\infty} X(s) e^{st} \, ds$$

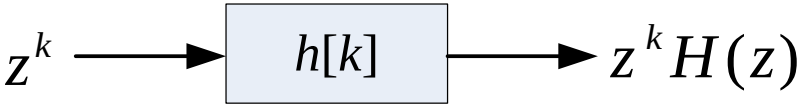
$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

$$y(t) = \frac{1}{2\pi j} \int_{\sigma-\infty}^{\sigma+\infty} X(s) H(s) e^{st} ds$$

$$Y(s) = H(s) X(s)$$

# Complex frequency domain analysis of signals and systems






$$z^k * h[k] = \sum_{n=-\infty}^{\infty} h[n] z^{(k-n)} = z^k H(z)$$

$$H(z) = \sum_{k=-\infty}^{\infty} h[k] z^{-k}$$

$$x[k] = \frac{1}{2\pi \mathrm{j}} \oint_C X(z)\, z^{k-1}\, \mathrm{d}z$$

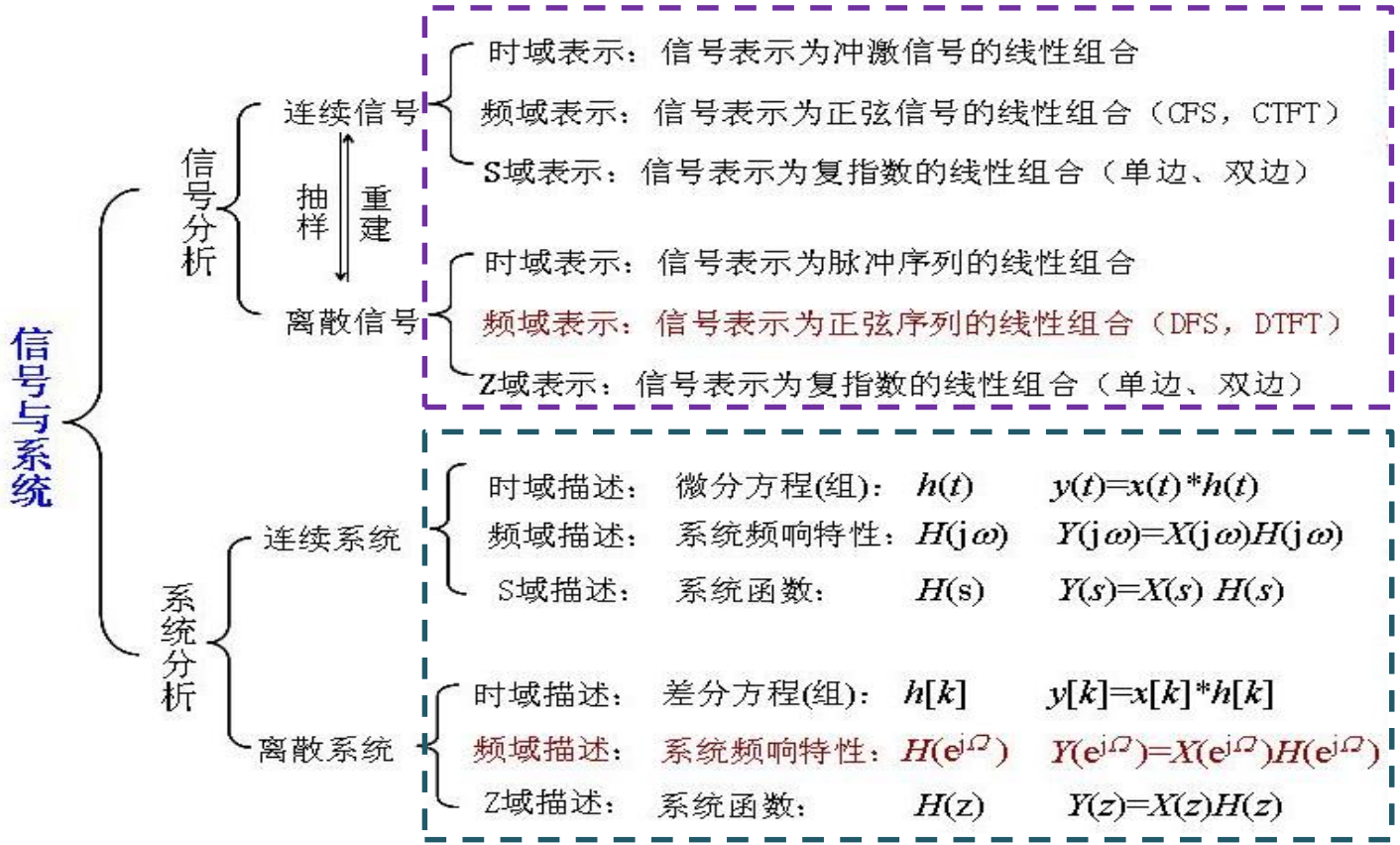
$$X(z) = \sum_{k=-\infty}^{\infty} x[k] z^{-k}$$

$$y[k] = \frac{1}{2\pi \mathrm{j}} \oint_C X(z) H(z) z^{k-1} \mathrm{d}z$$

$$Y(z) = H(z) X(z)$$

# The Teaching Connotation of Signal and System Course

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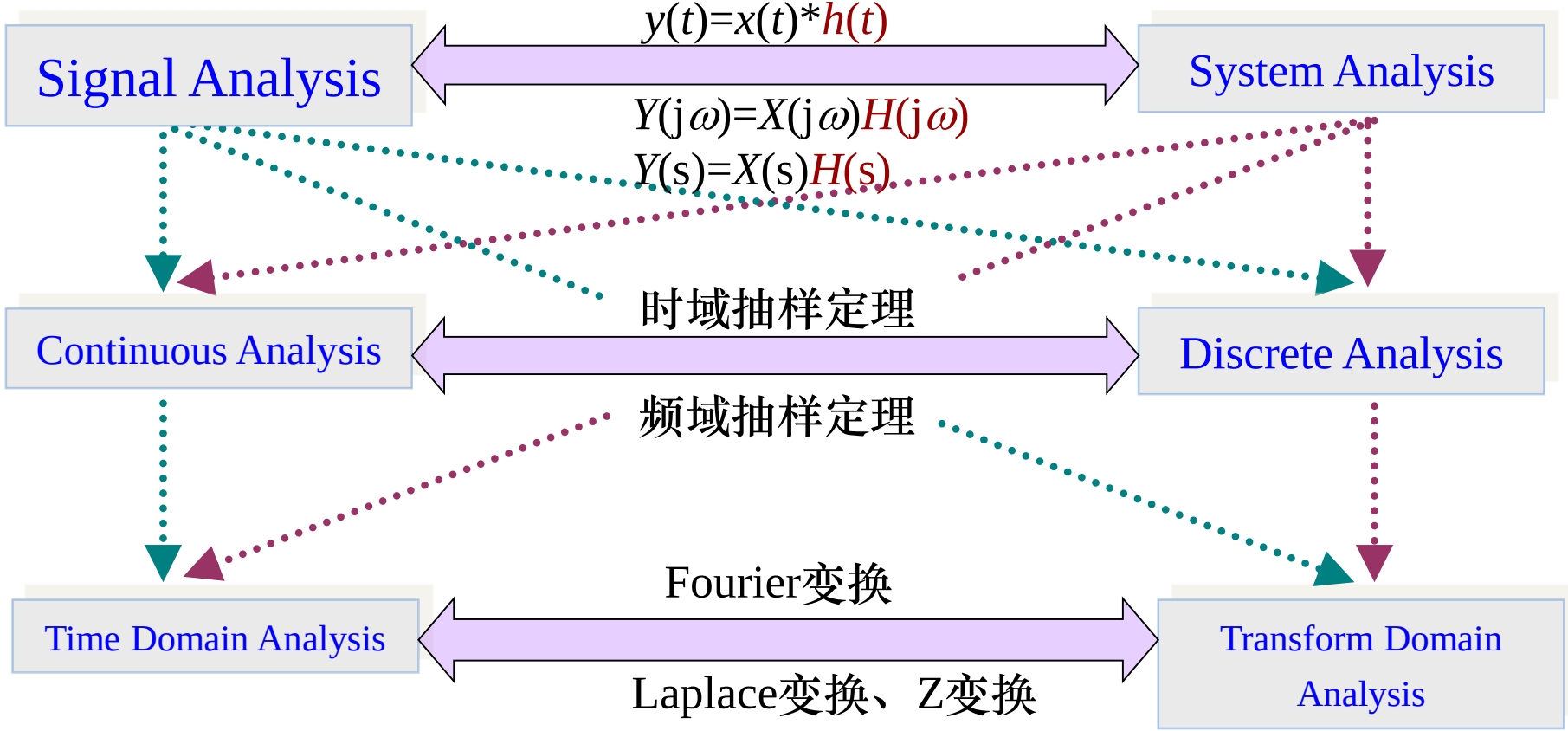



Signal Representation

System Description

# The Teaching Connotation of Signal and System Course

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# Why learn digital signal processing?

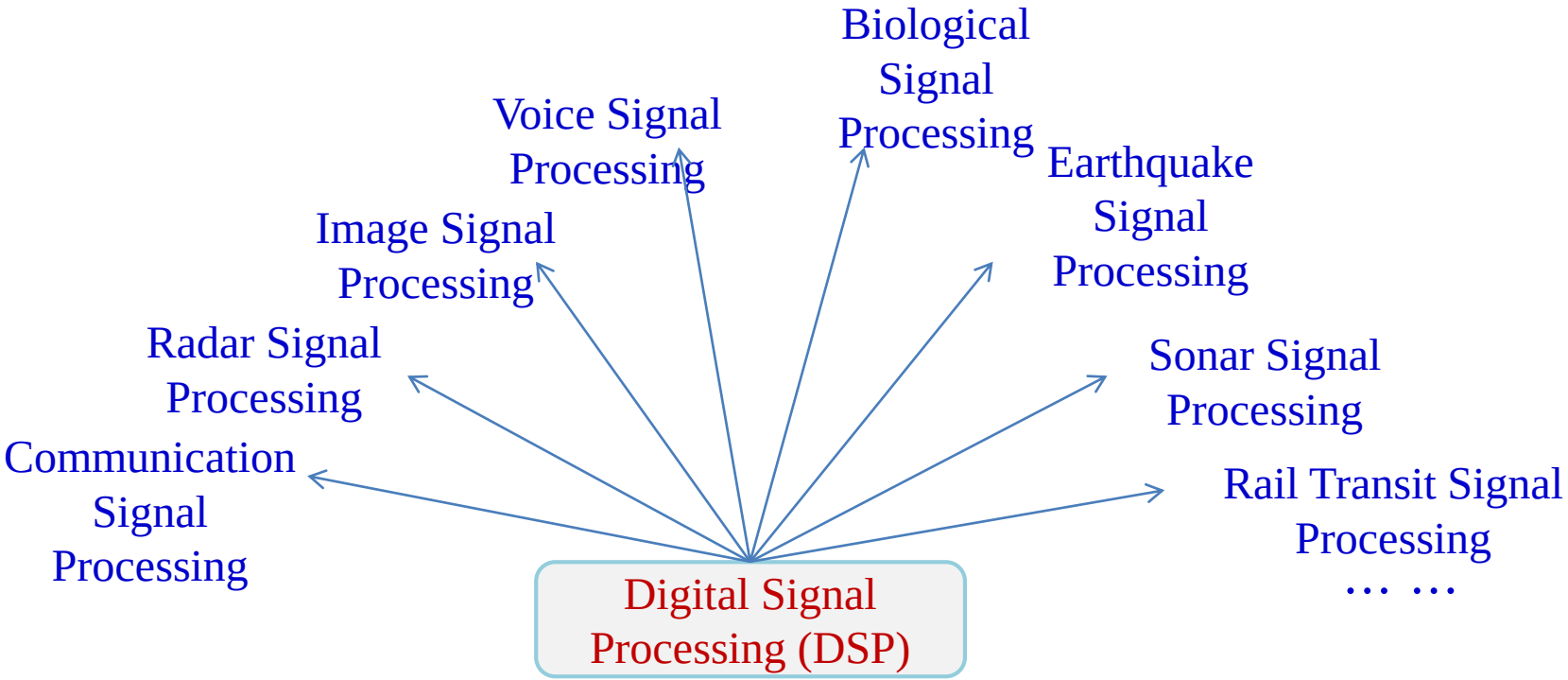


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| | |
|---|---|
| 1960s | With the emergence of algorithms such as FFT, Digital Signal Processing has risen |
| 1970s | Research on digital signal processing theories and algorithms, primarily for military applications |
| 1980s | DSP chips were introduced and rapidly developed, leading to the commercialization of digital products. |
| 1990s | Digital products have become widespread, with prices decreasing annually. |
| Since the 21st century | Digital technology has penetrated into various fields involving signal processing and daily life, changing our lives. |

# Why learn digital signal processing?

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# Why learn digital signal processing?

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- Digital technology is transforming our lives——The digitization of daily necessities

















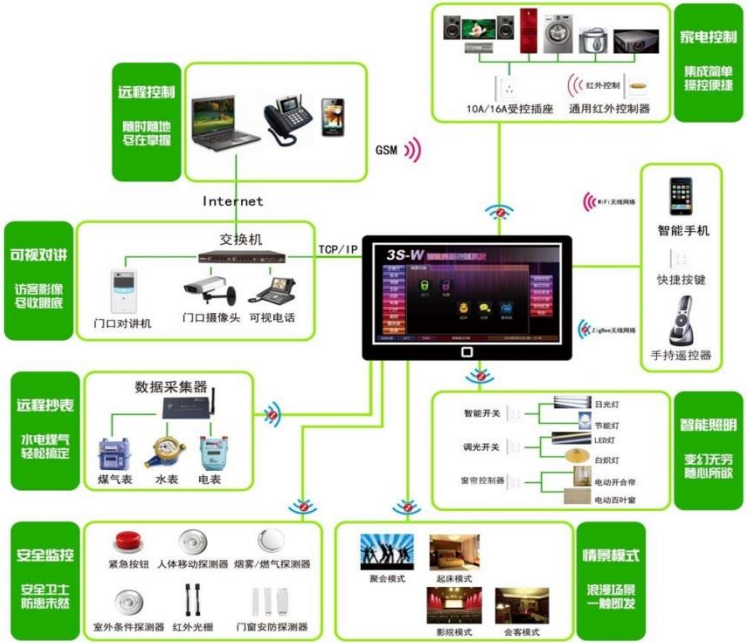





# Why learn digital signal processing?

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➢Digital technology is transforming our lives——The digitization of living environment

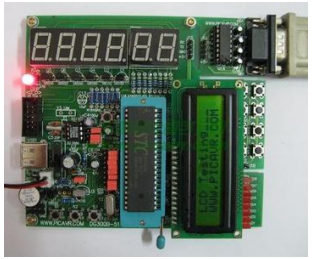



Digitalized Smart Home















Single-chip microcomputer、DSP (Digital Signal Processor)、Embedded Systems, etc.

Digital Kitchen

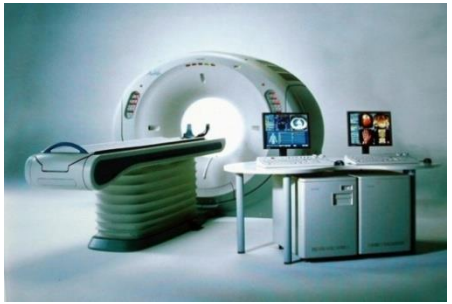
# Why learn digital signal processing?

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➢Digital technology is transforming our lives——The digitization of living environment

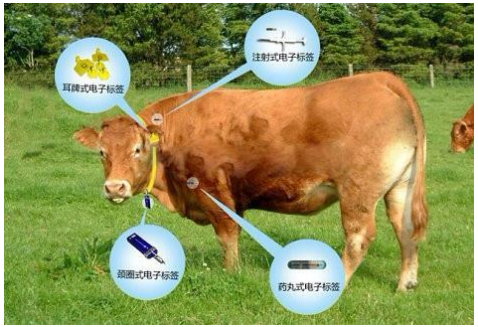




Digital hospital






Digital Farm

# Why learn digital signal processing?

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- Digital technology is transforming our lives——The digitization of lifestyle






Communication



Traffic









Shopping




Read




Teaching

# Why learn digital signal processing?

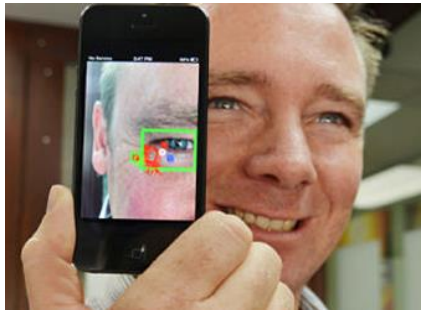
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➢Digital technology is transforming our lives——The digitization of life methods o









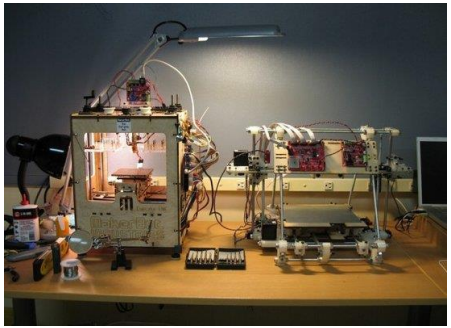


Music Download Platform









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Generally, binary is used, and any component with two stable states can be used to represent binary, so its basic unit circuit is simple， The precision requirements for each component in the circuit are not strict。

# The main characteristics of digital signals



- Easy to store
- Easy to transmit
- Easy to process

# The main characteristics of digital signals

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Easy to store

## Storage Methods

### Analog Storage



Video Tape



Tape




Camera Roll

Digital Storage
Advantages:
1. Large capacity
2. Good stability

### Digitl Storage




DVD



Hard Disk




SD Card

# The main characteristics of digital signals

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## Easy to store



Old Photos

《Durian Song》
Magnetic tape sound quality

## Stability




Digital Photos (Spatially Discrete)

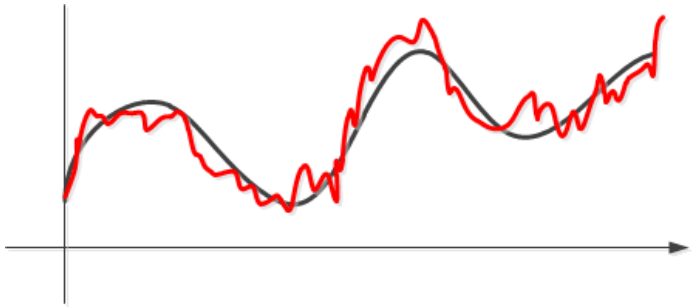
《Durian Song》
Magnetic tape sound quality

Digital storage method facilitates long-term stable storage of data

# The main characteristics of digital signals

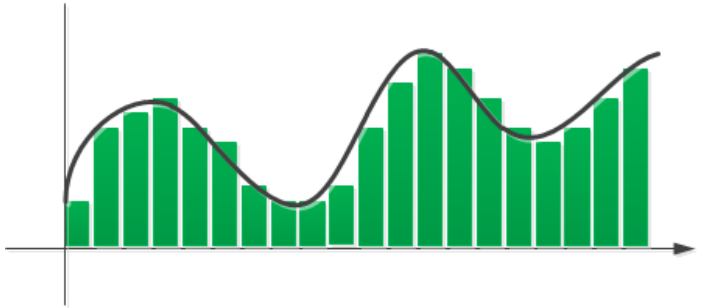
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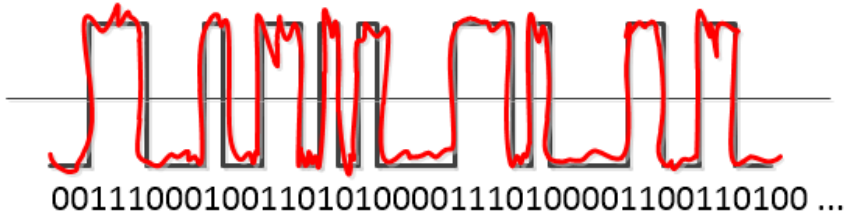
## Easy to Transmit



Noise interference in analog signal transmission

In digital transmission, signal amplitude is encoded using binary 0s and 1s, which provides stronger anti-interference capability compared to analog signals.






Noise interference in digital signal transmission

High Anti-interference Ability

# The main characteristics of digital signals



Easy to Transmit

High Anti-interference Ability

For example: the opening recording of China National Radio



Analog Signal Transmission



Digital Signal Transmission

Digital transmission effectively resists channel interference and provides great sound quality.

# The main characteristics of digital signals

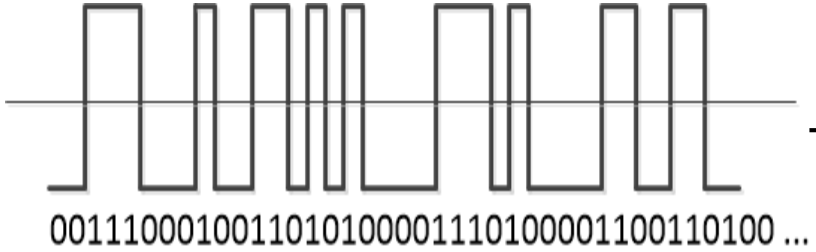
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Easy to Process

Signal Encryption



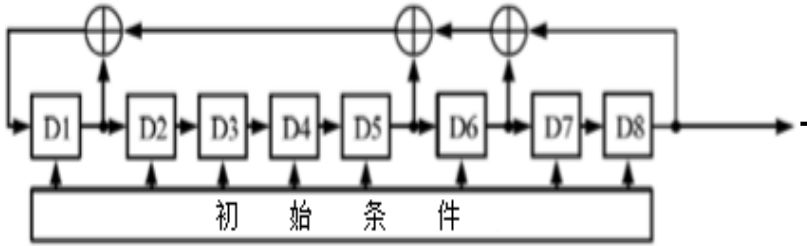
Voice Signal






Encrypted Voice

Pseudorandom code




High

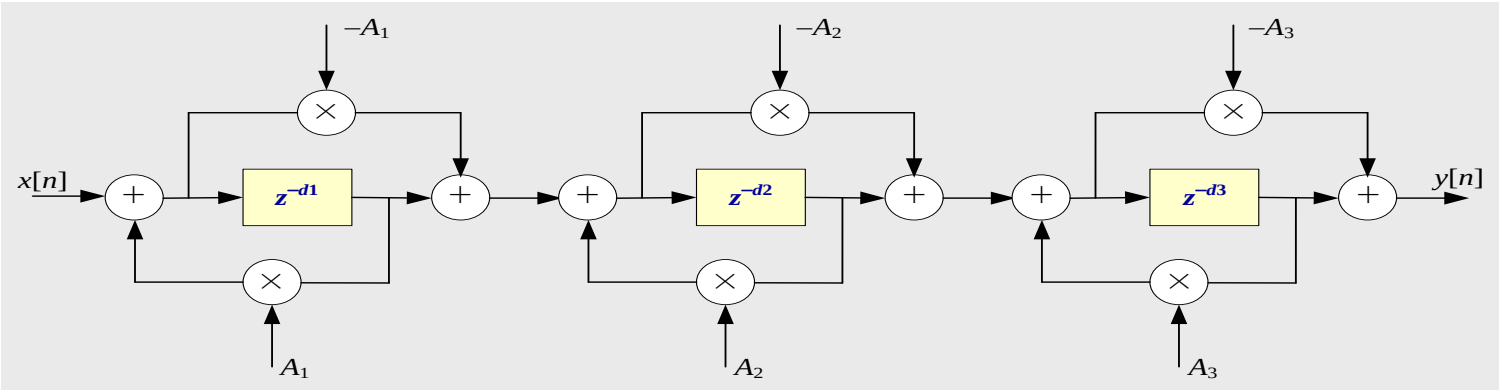
Confidentiality

# The main characteristics of digital signals

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## ➢Easy to Handle

The reverberation effect of music can be achieved through signal delay




Reverberation block diagram



Liu Huan, the original singer of "You and Me"

Processed audio signal

# The main characteristics of digital signals

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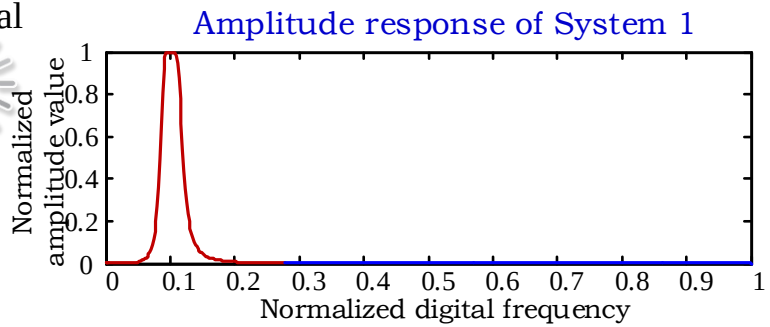
## ➢High Flexibility

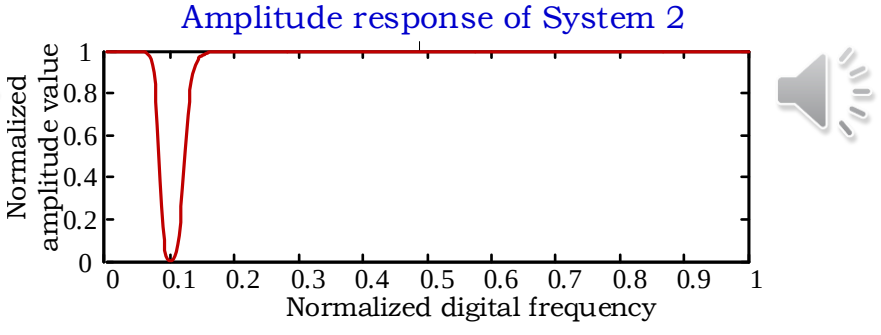
Changing the coefficients of the system function H(z) can realize different system functions.

$$H_1(z)=\frac{(0.0943\ -0.2828z^{-2}+0.2828z^{-4}-0.0943z^{-6})\times 10^{-3}}{1-5.5242z^{-1}+12.9897z^{-2}-16.6206z^{-3}+12.2014z^{-4}-4.8741z^{-5}+0.8288z^{-6}}$$

$$H_2(z)=\frac{0.8588\ -4.8975z^{-1}+11.8861z^{-2}-15.6940z^{-3}+11.8861z^{-4}-4.8975z^{-5}+0.8588z^{-6}}{1-5.4139z^{-1}+12.4787z^{-2}-15.6561z^{-3}+11.2736z^{-4}-4.4190z^{-5}+0.7375z^{-6}}$$

Original Signal



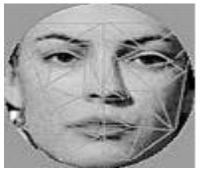



# The main characteristics of digital signals

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Easy to Process

Identify and Search



Collection and Positioning → Feature Extraction → Feature Matching → Search Results

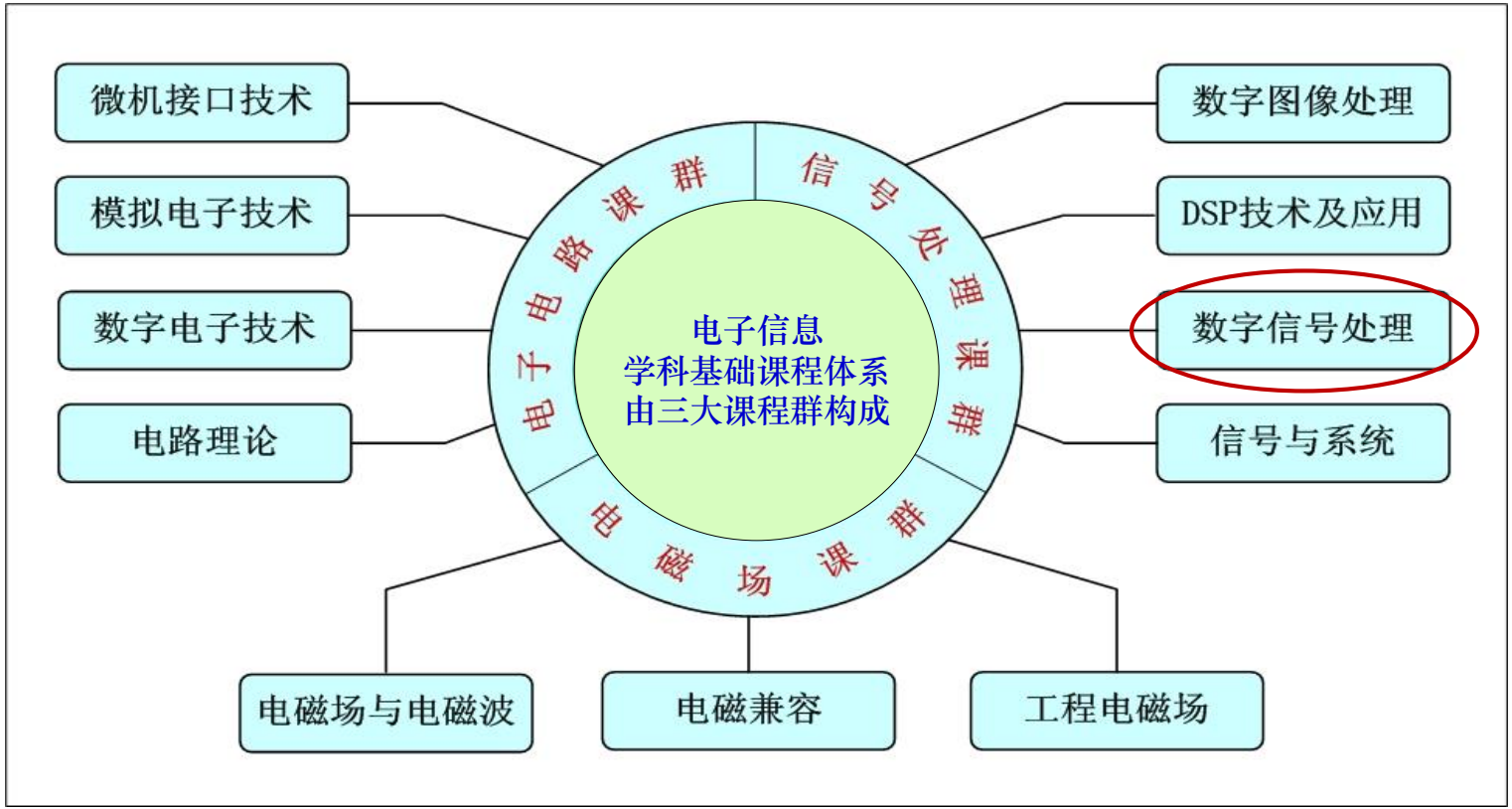






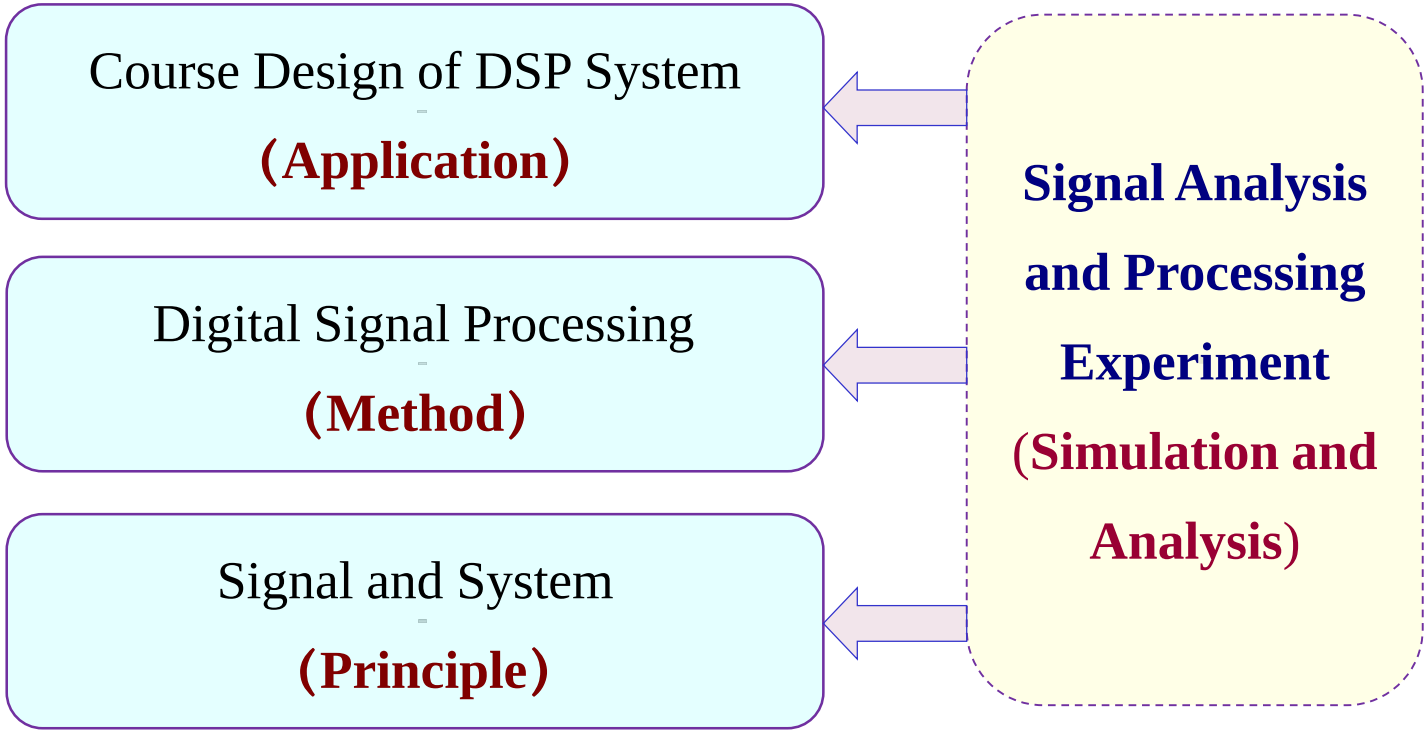

# Curriculum System for Electronic Information Majors

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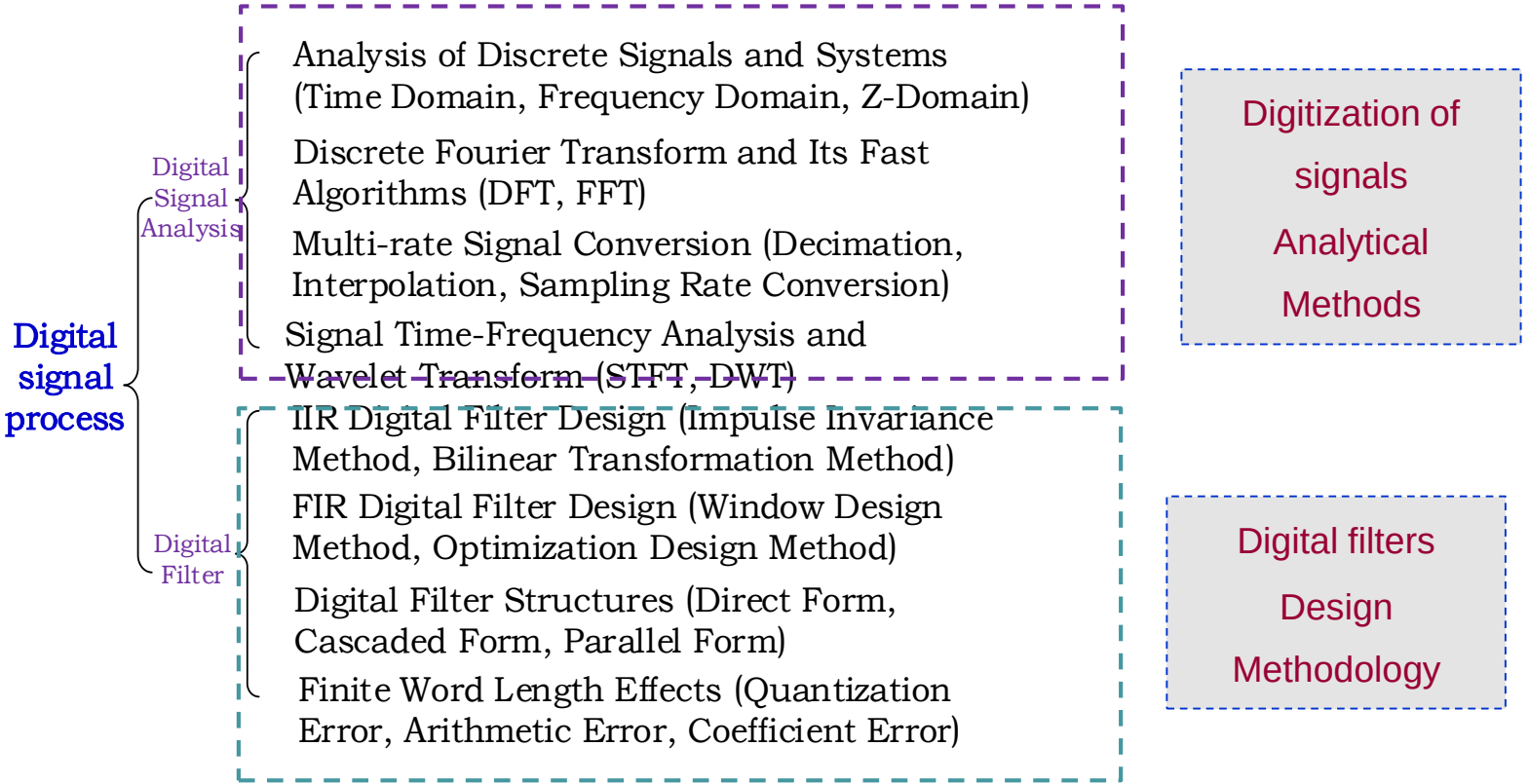
# The main characteristics of digital signals





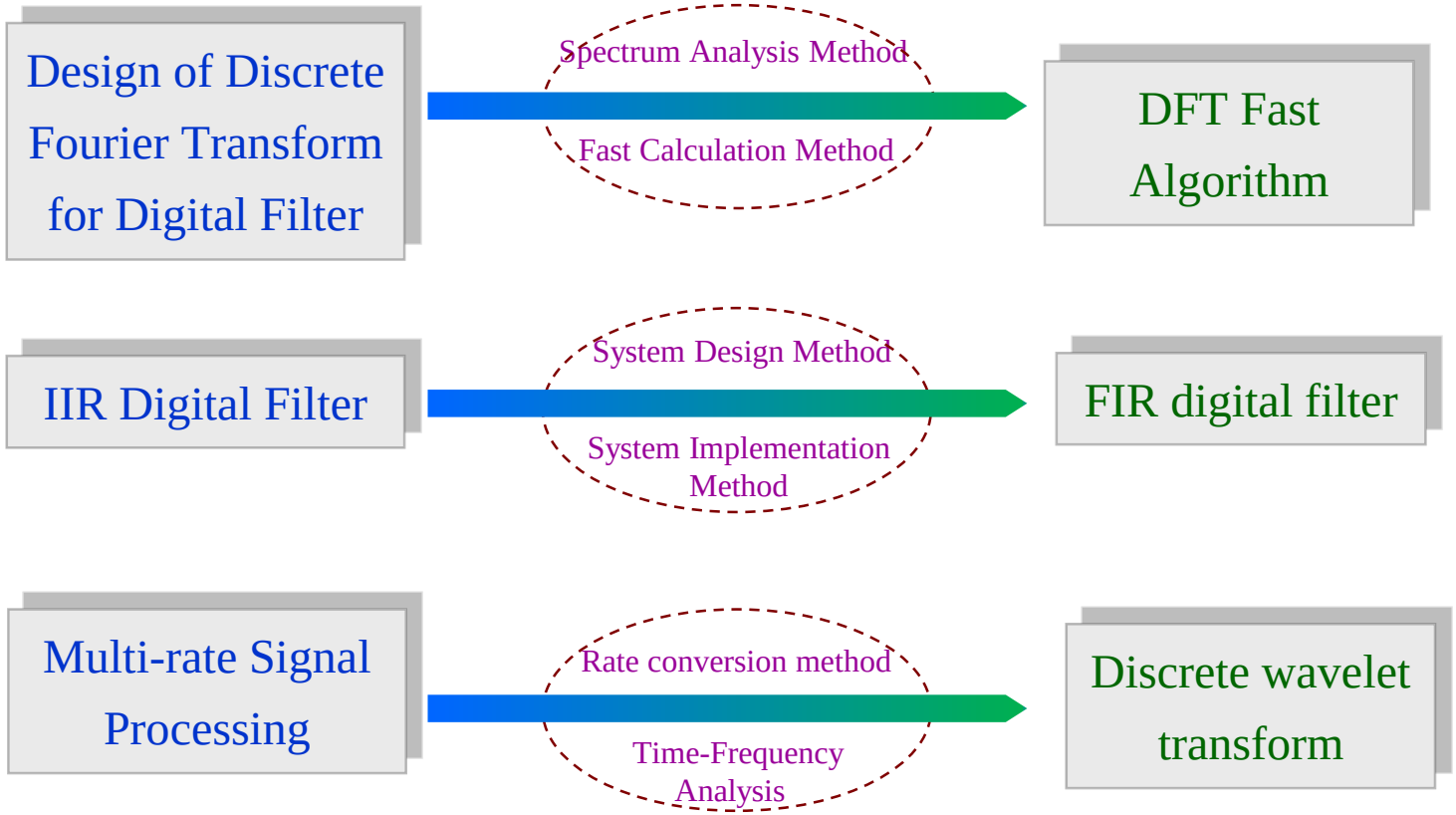
# Course Content of Digital Signal Processing





# Course Content of Digital Signal Processing





# Teaching Objectives of Digital Signal Processing Course



- Master the fundamental theories of digital signal processing

  Theories of Time-Domain and Transform-Domain Analysis, Sampling Theorem

- Master the fundamental methods of digital signal processing

  Digital analysis methods, filter design techniques, and fast algorithms

- Master the fundamental techniques of digital signal processing

  DSP software simulation and analysis techniques, DSP system development and application technologies

- Enhance practical ability to solve problems using DSP theories and techniques

  Applications in disciplinary fields, practical engineering applications, and interdisciplinary applications

# Teaching Resources for Digital Signal Processing Courses



- Main textbook:

普通高等教育“十四五”高等教育教材

《数字信号处理》（第4版）

陈后金等

高等教育出版社，2024

- Supporting textbook:

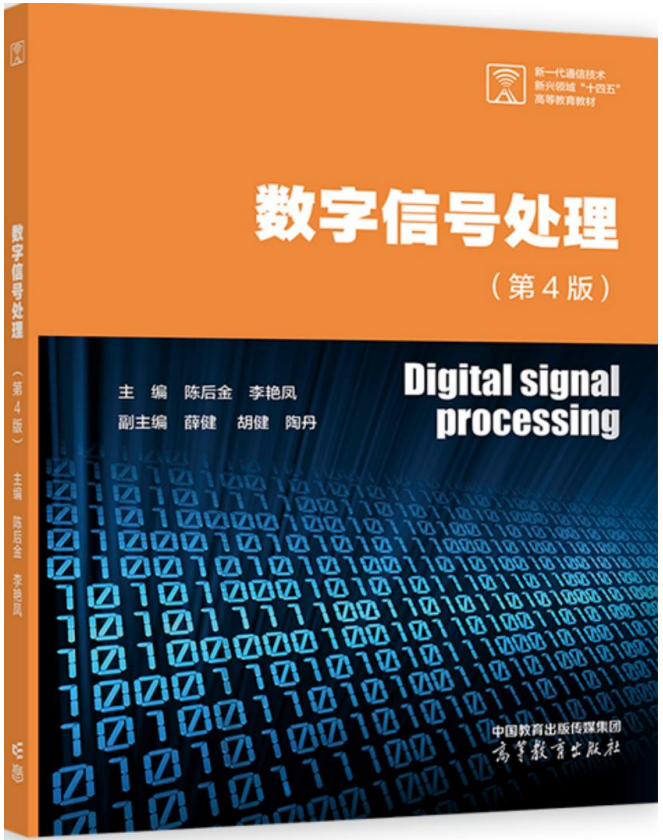
《数字信号处理学习指导及习题精解》

陈后金，薛健，胡健

高等教育出版社，2021

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# Thanks!

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude!